Getting in Sync: Health and Digital Literacy in Patient Deep Brain Stimulation Device Use Rita A. Shewbridge MS1, Helena M. Mentis PhD1, Courtney D. Pharr2, Sharon Powell RN3, Paul Fishman MD3, Melissa Armstrong MD3, Lisa Shulman, MD3 1University of Maryland Baltimore County, Baltimore, MD; 2Clark University, Worcester, MA; 3University of Maryland School of Medicine, Baltimore, MD Abstract With the growing number of mobile health devices, a description of the factors that lead to adoption and use of these personal health devices is needed. We observed and recorded 11 routine deep brain stimulation (DBS) program- ming appointments. We found an overall lack of health literacy surrounding DBS as well as a lack of digital literacy in patients' ability to use their DBS Patient Programmer to modify their DBS settings. These two factors led to pa- tient disuse of their programming device outside of the clinical setting. This paper's contribution is to present this relationship and suggest ways we can better facilitate patients in gaining both health and digital literacy. Introduction A turn towards a patient-centered care model has prompted an interest in providing patients with devices to monitor their health, adjust treatments accordingly, and share this data with their clinicians. In essence, putting power in the patient's hand to give them agency over their own care. However, the context around device usage could be a signif- icant factor in evaluating the efficacy of these devices in actually achieving patient engagement or empowerment. Most significant in this regard is the importance of health literacy and numeracy in understanding one's disease and the relationship between symptoms and treatments [1]. In addition, as further technologically advanced devices are developed, a patient's ability to understand and use the technology plays a greater role in patient engagement [2]. In the following study, we describe the particular case of deep brain stimulation (DBS) in Parkinson's disease (PD) symptom management. In this instance, patients are provided with a device to modify the brain stimulation parame- ters to better address symptoms throughout the day. As we show in our findings, low health literacy and digital liter- acy often limit and occasionally altogether deter patients from modifying their stimulation device parameters outside of the clinical setting. This presents a real challenge in (1) presenting patients with health devices to maintain control over their own quality of life and (2) engaging patients in guiding clinicians in making further treatment changes that provide therapeutic benefit. Our findings lead us to make design recommendations for the development of visu- alizations to explicate the relationship between DBS adjustments and symptomatic outcomes in order to facilitate patient engagement through increased health literacy and digital literacy. Background There has been a shift from the traditional clinical care model, where clinicians are solely in charge of the care of the patient, to a patient-centered care model that is more focused on patients being engaged in their own medical deci- sion-making and treatment [3]. Patient engagement has been associated with a decreased number of annual visits for specialty care and fewer hospitalizations and diagnostic tests [4]. Prior work has shown that health literacy is important for patient engagement as patients can then better manage their health, make more informed decisions, and increase communication with their doctors [1]. Health literacy al- lows chronically and progressively ill patients to better "obtain, process, and understand" their health conditions and assists them in making more informed decisions surrounding their care [5]. With low health literacy, patients often lack the skills to effectively communicate important information, such as symptoms or other health concerns, to their doctors, and they frequently feel that their physicians fail to explain health related information in a manner that they can understand [6]. This, in turn, can lead to poor compliance when it comes to taking medications or following a treatment plan, and as a result, worse health outcomes [6]. Likewise, digital literacy, the ability to appropriately use digital tools to access and manage digital resources [7], has also been shown to be a fundamental factor in patient engagement primarily with regards to searching for online health information [2]. For instance, in a 2013 study examining the relationship between functional health literacy, self-efficacy, social support, and empowerment in individuals living with Parkinson's, nearly 50% of patients indi- cated that the Internet was the most common source they used to find health information relating to their disease [8].